

Question 2 Clustering: Full Trait Model

This notebook clusters dog breeds using the full set of numeric AKC trait scores, then compares those clusters to popularity tier and AKC group without regrouping the traits.

Cluster Summary on AKC Groups

Overall Metrics
Overall silhouette score: 0.1
Rand Index: 0.77
Average radius / average intercluster distance: 1.27

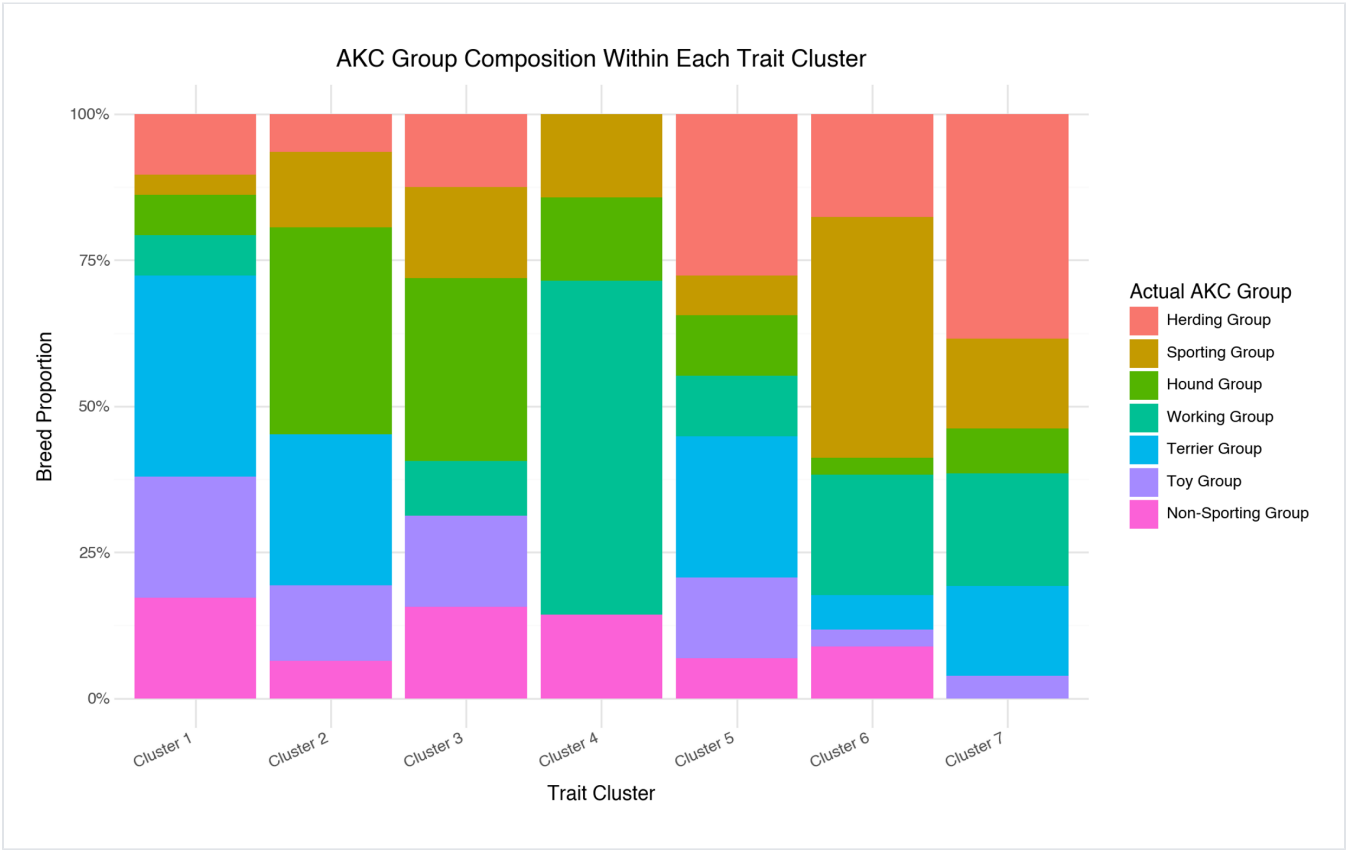
Trait Cluster Metric	Cluster 1	Cluster 2	Cluster 3	Cluster 4	Cluster 5
Cluster Size	29	31	32	21	29
Most Common AKC Group	Terrier Group	Hound Group	Hound Group	Working Group	Herding Group
Most Common AKC Group Count	10	11	10	12	8
Predicted AKC Group	Terrier Group	Hound Group	Hound Group	Working Group	Herding Group
Distance To Predicted AKC Group	1.27	1.48	1.79	1.56	1.62
AKC Rand Index	0.68	0.67	0.66	0.75	0.65
Silhouette Score	0.16	0.06	0.09	0.14	0.06
Radius	3.82	3.55	3.76	4.21	3.87

Cluster Centroids vs AKC Group Traits Averages

	Cluster 1	Cluster 2	Cluster 3	Cluster 4	Cluster 5	Cluster 6	Cluster 7
Trait							
Affectionate With Family	4.90	4.45	4.50	4.52	4.97	4.91	3.12
Good With Young Children	3.07	3.61	4.81	3.43	3.79	4.88	3.04
Good With Other Dogs	2.86	3.68	4.66	3.10	3.38	3.88	2.88
Shedding Level	2.24	1.81	2.84	3.05	2.72	3.00	2.54
Coat Grooming Frequency	2.52	1.81	2.50	1.81	2.24	2.15	2.81
Drooling Level	1.45	1.19	1.75	3.57	1.52	2.12	1.23
Openness To Strangers	3.17	3.06	3.78	3.05	4.07	4.06	2.92
Playfulness Level	3.45	3.48	3.62	3.10	4.21	4.26	3.15
Watchdog/Protective Nature	4.31	3.03	2.91	4.48	4.21	3.56	3.92
Adaptability Level	3.76	3.61	3.94	3.48	4.10	4.21	3.15
Trainability Level	3.45	3.48	3.66	3.62	4.17	4.74	3.73
Energy Level	3.24	3.55	3.53	3.19	4.34	4.32	3.85
Barking Level	3.83	2.39	3.75	2.10	4.00	2.85	2.81
Mental Stimulation Needs	3.38	3.48	3.34	3.43	4.17	4.26	3.50

AKC Group	Herding Group	Hound Group	Non-Sporting Group	Sporting Group	Terrier Group	Toy Group	Working Group
Trait							
Affectionate With Family	4.18	4.25	4.85	4.64	4.68	4.76	4.44
Good With Young Children	3.70	3.78	4.00	4.12	3.97	3.57	3.91
Good With Other Dogs	3.36	4.22	3.40	3.91	2.94	3.76	3.22
Shedding Level	2.82	2.50	2.50	2.70	2.16	2.33	2.97
Coat Grooming Frequency	2.42	1.66	2.50	2.33	2.35	2.57	2.22
Drooling Level	1.55	1.75	1.65	2.12	1.29	1.10	2.66
Openness To Strangers	3.12	3.22	3.80	3.73	3.65	3.81	3.31
Playfulness Level	3.64	3.38	3.80	3.85	3.68	3.81	3.53
Watchdog/Protective Nature	3.91	3.06	3.80	3.18	3.94	3.62	4.53
Adaptability Level	3.70	3.53	3.90	3.94	3.81	4.14	3.62
Trainability Level	4.24	3.25	3.60	4.64	3.52	3.67	3.91
Energy Level	4.06	3.72	3.50	4.09	3.61	3.29	3.69
Barking Level	3.39	3.31	2.90	2.73	3.55	3.33	2.78
Mental Stimulation Needs	4.00	3.62	3.40	3.82	3.61	3.29	3.72

How well do trait-based clusters line up with AKC Groups?



Prediction Output

This table saves each breed's assigned cluster with its average rank, popularity tier, and AKC group.

	Breed	Trait Cluster	Predicted AKC Group	Actual AKC Group	Distance To Predicted AKC Group	Correct AKC Prediction
0	Affenpinscher	Cluster 7	Herding Group	Toy Group	1.90	False
1	Afghan Hound	Cluster 7	Herding Group	Hound Group	1.90	False
2	Airedale Terrier	Cluster 7	Herding Group	Terrier Group	1.90	False
3	Akita	Cluster 7	Herding Group	Working Group	1.90	False
4	Alaskan Malamute	Cluster 7	Herding Group	Working Group	1.90	False
...	...	...	...	...	...	...
197	Wire Fox Terrier	Cluster 5	Herding Group	Terrier Group	1.62	False
198	Wirehaired Pointing Griffon	Cluster 6	Sporting Group	Sporting Group	1.25	True
199	Wirehaired Vizsla	Cluster 6	Sporting Group	Sporting Group	1.25	True
200	Xoloitzcuintli	Cluster 2	Hound Group	Non-Sporting Group	1.48	False
201	Yorkshire Terrier	Cluster 5	Herding Group	Toy Group	1.62	False

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